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ETE 3351
Lab 10

Lab 10: FM Demodulation

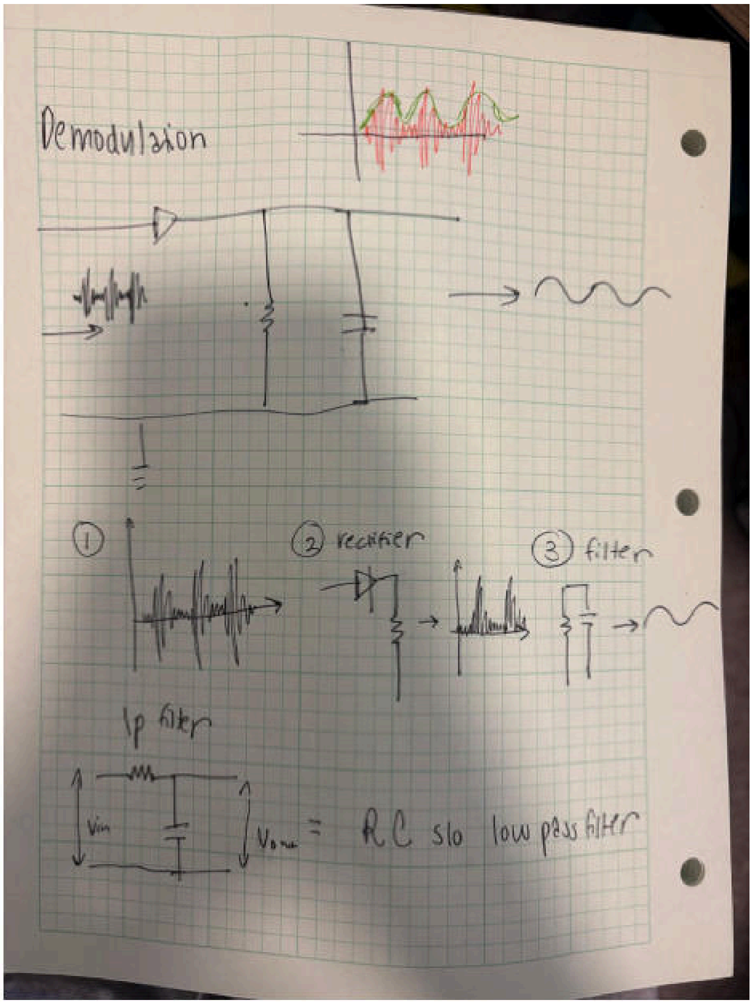
Introduction

In AM (Amplitude Modulation we usually only need one simple type of circuit to recover the original message called a diode detector (see image 1). However for FM (Frequency Modulation) there are several different detector types because the information is carried in changes in frequency rather than changes in amplitude. In this lab two FM demodulation methods are studied: the slope detector and the Phase-Locked Loop (PLL).

FM signals are commonly generated using a Voltage Controlled Oscillator (VCO) where an input voltage causes the output frequency to increase or decrease. To demodulate FM this process must be reversed by converting frequency changes back into voltage changes. Before demodulation the FM signal is passed through a limiter amplifier which removes any unwanted amplitude variations caused by noise. This step is acceptable because in FM the amplitude does not carry information, so clipping or limiting the signal does not affect the message. However, trimming the amplitude is absolutely critical to not do for AM because information will be lost.

The slope detector is the simplest FM demodulator examined by us in this lab. It uses a band-pass filter with a sloped frequency response, meaning the filter gain changes with frequency. When the FM signal shifts above or below the carrier frequency it lands on different parts of the slope, causing the output amplitude to increase or decrease. This effectively converts frequency variations into amplitude variations, creating a signal that looks similar to an AM signal, often called a *pseudo*-AM signal. Once this conversion is complete, a standard diode detector (similar to the one used in AM demodulation in image 1) is used to recover the original audio signal.

The PLL is a more advanced FM demodulation technique. It consists of a phase detector, a low-pass filter, and a VCO. The phase detector compares the incoming FM signal with the VCO output and produces an error signal based on the difference between them. This error signal is filtered and used to adjust the VCO frequency so that it tracks the input signal. When the PLL is locked, the control voltage driving the VCO follows the original modulation, allowing the message signal to be recovered accurately. This lab demonstrates how both basic and advanced circuits can be used to demodulate FM signals by converting frequency changes back into usable information.



IMG 1.

Step1: Generate an FM Signal suggested values.

Amplitude 5Vp-p, frequency 15kHz: Modulation frequency 100Hz, deviation 5kHz, Sine Shape
Use example 9-2 in the book to predict and determine the following values

Use example 9-2 in the book to predict and determine the following values

EXAMPLE 9–2

An FM signal expressed as $v_{FM} = 1000 \cos(2\pi 10^7 t + 0.5 \cos 2\pi 10^4 t)$ is measured in a 50-Ω antenna. Determine the following:

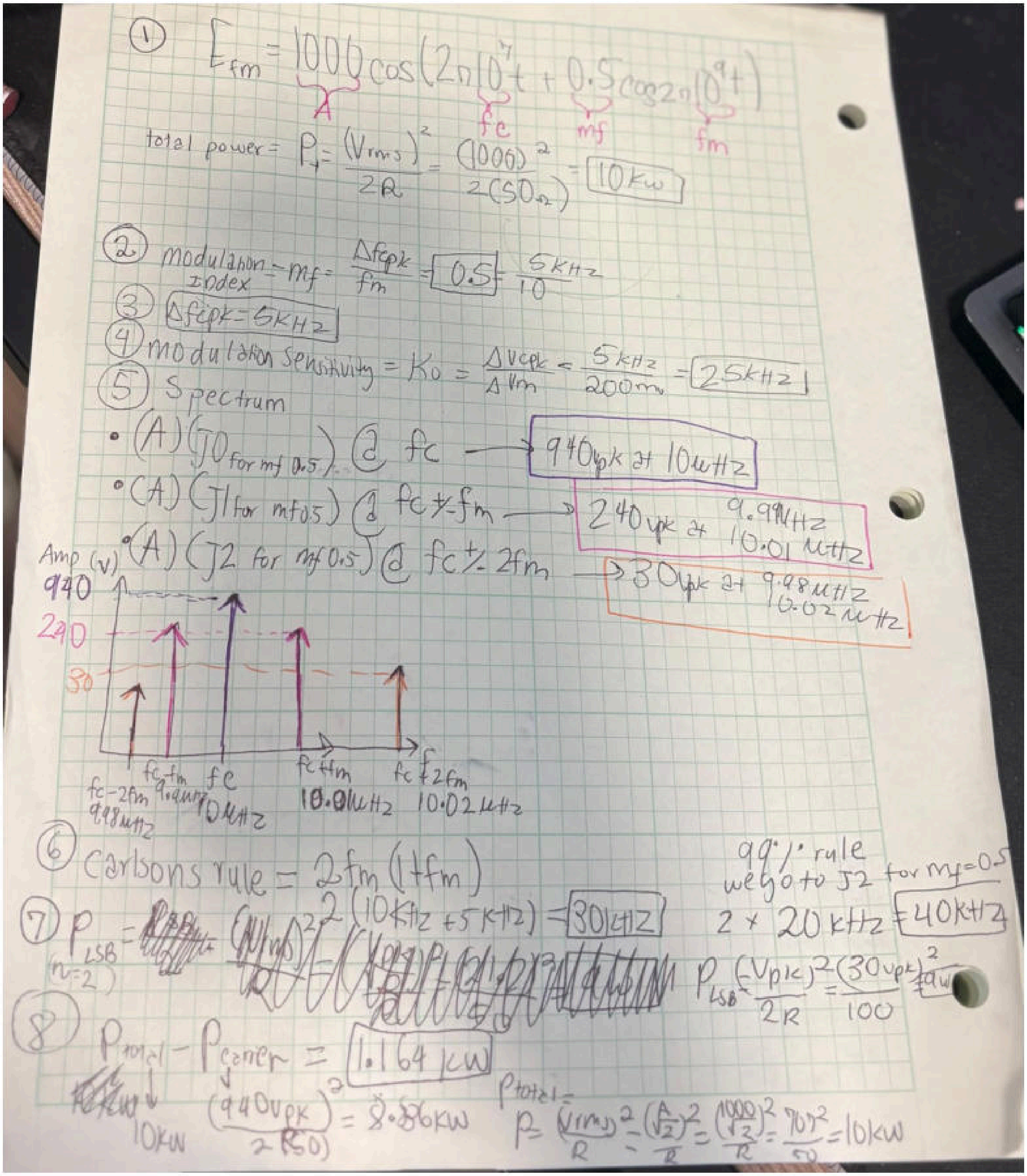
1. Total power
2. Modulation index
3. Peak frequency deviation
4. Modulation sensitivity if 200 mV pk is required to achieve part 3
5. Spectrum
6. Bandwidth (99%) and approximate circuit bandwidth by Carson's rule
7. Power in the smallest sideband (just one sideband) of the 99% bandwidth
8. Total information power

IMG 2 . *Problem 9-2*

TABLE 9–1 Bessel Functions of the First Kind, $J_n(m_f)$

m_f	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}	J_{11}	J_{12}	J_{13}	J_{14}
0.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0.25	0.98	0.12	—	—	—	—	—	—	—	—	—	—	—	—	—
0.5	0.94	0.24	0.03	—	—	—	—	—	—	—	—	—	—	—	—
1.0	0.77	0.44	0.11	0.02	—	—	—	—	—	—	—	—	—	—	—
1.5	0.51	0.56	0.23	0.06	0.01	—	—	—	—	—	—	—	—	—	—
2.0	0.22	0.58	0.35	0.13	0.03	—	—	—	—	—	—	—	—	—	—
2.4	0	0.52	0.43	0.20	0.06	0.02	—	—	—	—	—	—	—	—	—
2.5	−0.05	0.50	0.45	0.22	0.07	0.02	0.01	—	—	—	—	—	—	—	—
3.0	−0.26	0.34	0.49	0.31	0.13	0.04	0.01	—	—	—	—	—	—	—	—
4.0	−0.40	−0.07	0.36	0.43	0.28	0.13	0.05	0.02	—	—	—	—	—	—	—
5.0	−0.18	−0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02	—	—	—	—	—	—
6.0	0.15	−0.28	−0.24	0.11	0.36	0.36	0.25	0.13	0.05	0.02	—	—	—	—	—
7.0	0.30	0.00	−0.30	−0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02	—	—	—	—
8.0	0.17	0.23	−0.11	−0.29	−0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03	—	—	—
9.0	−0.09	0.25	0.14	−0.18	−0.27	−0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01	—
10.0	−0.25	0.05	0.25	0.06	−0.22	−0.23	−0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03	0.01

IMG 3. *Bessel table used to solve 9-2*



① $E_{fm} = \underbrace{1000}_{A} \cos(2\pi \underbrace{10^7}_{fc} t + \underbrace{0.5}_{mf} \cos 2\pi \underbrace{10^4}_{fm} t)$

total power $= P_T = \frac{(V_{rms})^2}{2R} = \frac{(1000)^2}{2(50\Omega)} = \boxed{10 kW}$

② modulation index $= m_f = \frac{\Delta f_{cpk}}{f_m} = \boxed{0.5} = \frac{5 kHz}{10}$

③ $\boxed{\Delta f_{cpk} = 5 kHz}$

④ modulation sensitivity $= K_0 = \frac{\Delta V_{cpk}}{\Delta V_m} = \frac{5 kHz}{200 mv} = \boxed{25 kHz}$

⑤ Spectrum

- (A)(J0 for mf 0.5) @ fc → $\boxed{940 vpk \text{ at } 10 MHz}$
- (A)(J1 for mf 0.5) @ fc +/- fm → $\boxed{240 vpk \text{ at } 9.99 MHz,\ 10.01 MHz}$
- (A)(J2 for mf 0.5) @ fc +/- 2fm → $\boxed{30 vpk \text{ at } 9.98 MHz,\ 10.02 MHz}$

⑥ Carlsons rule $= 2f_m(1+f_m)$ $= 2(10kHz + 5kHz) = \boxed{30 kHz}$

99% rule we go to J2 for mf = 0.5

$2 \times 20 kHz = \boxed{40 kHz}$

⑦ P_{LSB} (n=2) $P_{LSB} = \frac{(V_{pk})^2}{2R} = \frac{(30 v_{pk})^2}{100} = \boxed{9 W}$

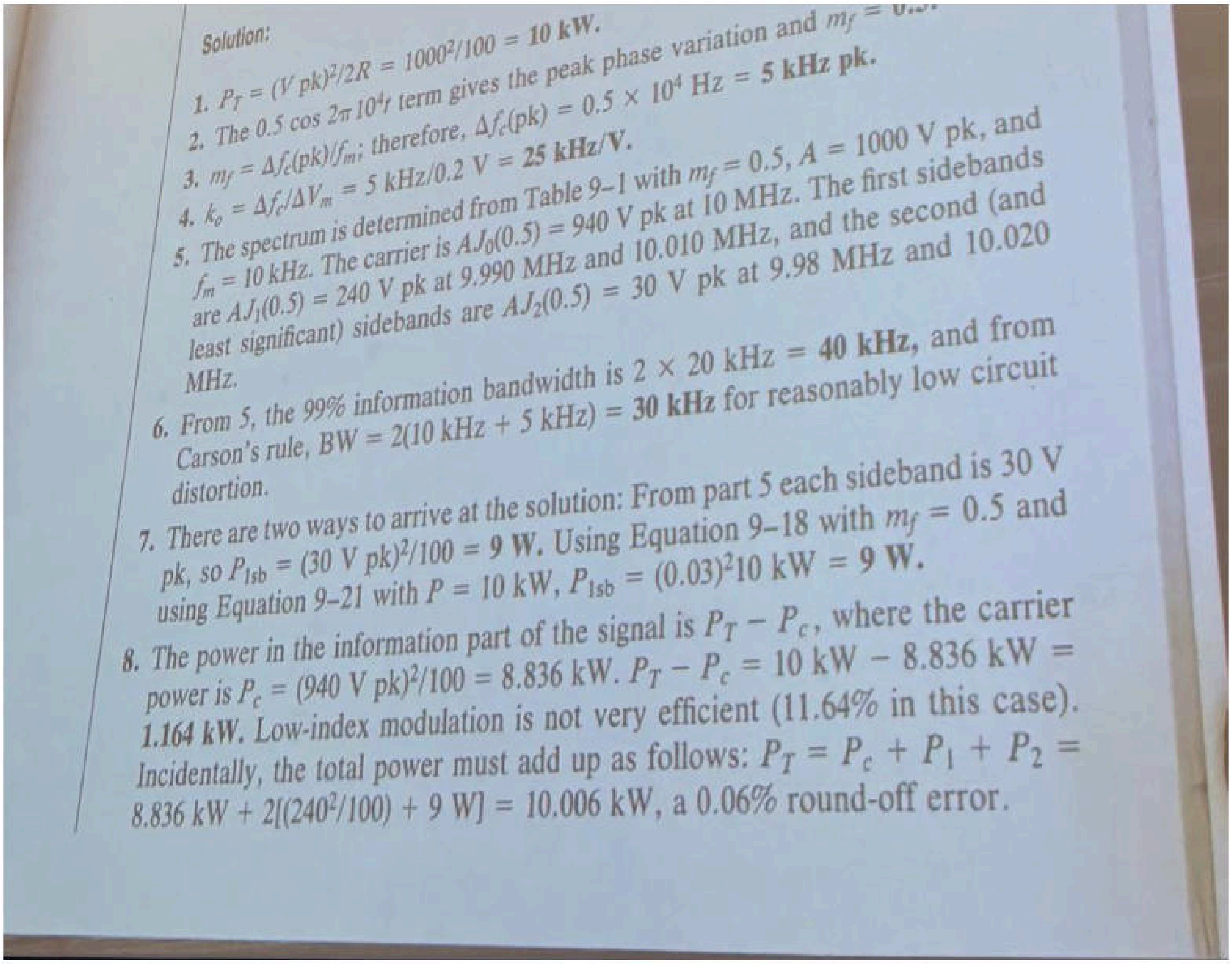
⑧ $P_{total} - P_{carrier} = \boxed{1.164 kW}$

$P_{total} \downarrow$ 10kW

$P_{carrier} \downarrow$ $\frac{(940 v_{pk})^2}{2(50)} = 8.836 kW$

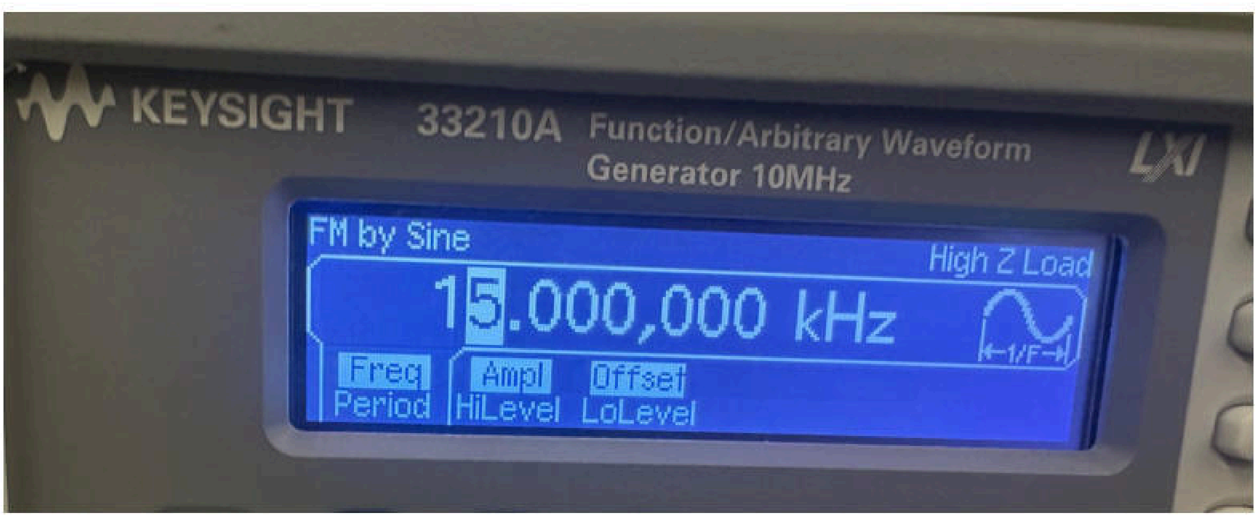
$P_{total} =$ $P = \frac{(V_{rms})^2}{R} = \frac{(\frac{A}{\sqrt{2}})^2}{R} = \frac{(\frac{1000}{\sqrt{2}})^2}{R} = \frac{707^2}{50} = 10 kW$

IMG 4 . *My work for problem 9-2*

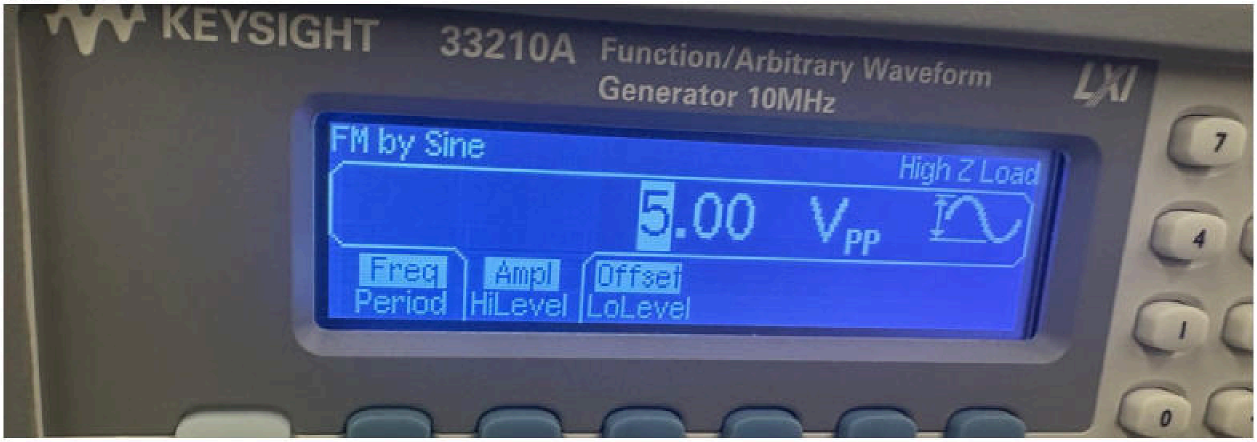

Solution:

1. $P_T = (V\ pk)^2/2R = 1000^2/100 =$ **10 kW.**
2. The 0.5 cos $2\pi 10^4 t$ term gives the peak phase variation and $m_f = 0.5$.
3. $m_f = \Delta f_c(pk)/f_m$; therefore, $\Delta f_c(pk) = 0.5 \times 10^4$ Hz = **5 kHz pk.**
4. $k_o = \Delta f_c/\Delta V_m = 5$ kHz/0.2 V = **25 kHz/V.**
5. The spectrum is determined from Table 9–1 with $m_f = 0.5$, $A = 1000$ V pk, and $f_m = 10$ kHz. The carrier is $AJ_0(0.5) = 940$ V pk at 10 MHz. The first sidebands are $AJ_1(0.5) = 240$ V pk at 9.990 MHz and 10.010 MHz, and the second (and least significant) sidebands are $AJ_2(0.5) = 30$ V pk at 9.98 MHz and 10.020 MHz.
6. From 5, the 99% information bandwidth is 2 × 20 kHz = **40 kHz,** and from Carson's rule, BW = 2(10 kHz + 5 kHz) = **30 kHz** for reasonably low circuit distortion.
7. There are two ways to arrive at the solution: From part 5 each sideband is 30 V pk, so $P_{1sb} = (30\text{ V pk})^2/100 =$ **9 W.** Using Equation 9–18 with $m_f = 0.5$ and using Equation 9–21 with $P = 10$ kW, $P_{1sb} = (0.03)^2 10$ kW = **9 W.**
8. The power in the information part of the signal is $P_T - P_c$, where the carrier power is $P_c = (940\text{ V pk})^2/100 = 8.836$ kW. $P_T - P_c = 10$ kW − 8.836 kW = **1.164 kW.** Low-index modulation is not very efficient (11.64% in this case). Incidentally, the total power must add up as follows: $P_T = P_c + P_1 + P_2 =$ 8.836 kW + $2[(240^2/100) + 9\text{ W}]$ = 10.006 kW, a 0.06% round-off error.

IMG 5. *Answers for 9-2, which match my solution in the previous image.*



IMG 6 *.fc*



IMG 7 . *Amplitude*

IMG 8 . *Deviation*

IMG 9 . *Modulation frequency*

Step2: Setup the filter for Demodulation i.e. Slope detection, see figure 9-24 from book.

Filter (use the Krohn-Hite on the bench) and the high-pass characteristic (left side control) set it to ~18kHz which is above the carrier of 15kHz, the low-pass control on the right side should be set to its maximum or 1MHz

IMG 10 . *filter setting*

IMG 11 . *filter setting*

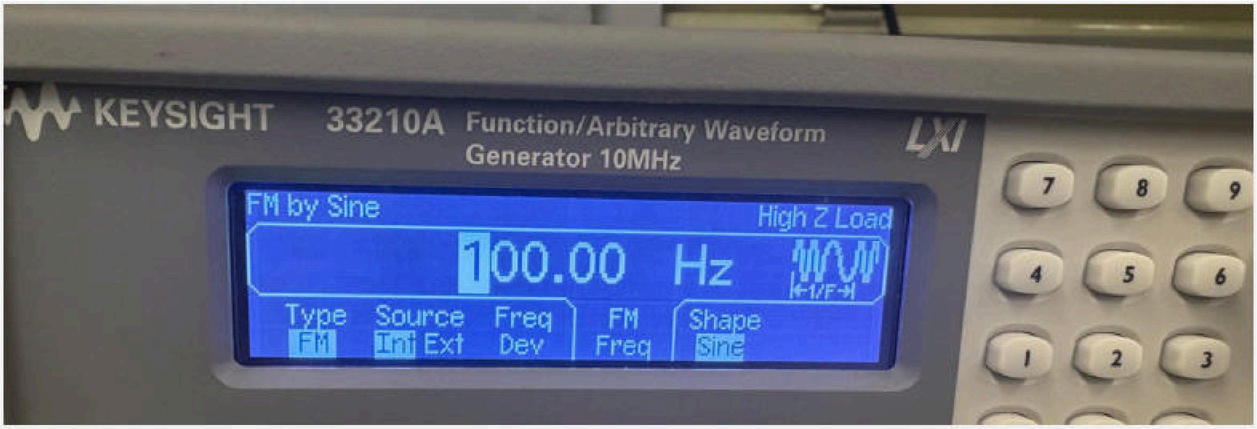
Step3: Review the filter output with an FM input signal.

The slope-detector output should look like a pseudo-AM signal in that frequencies above the rest frequency will be higher voltages while frequencies below the rest frequency will be lower voltages

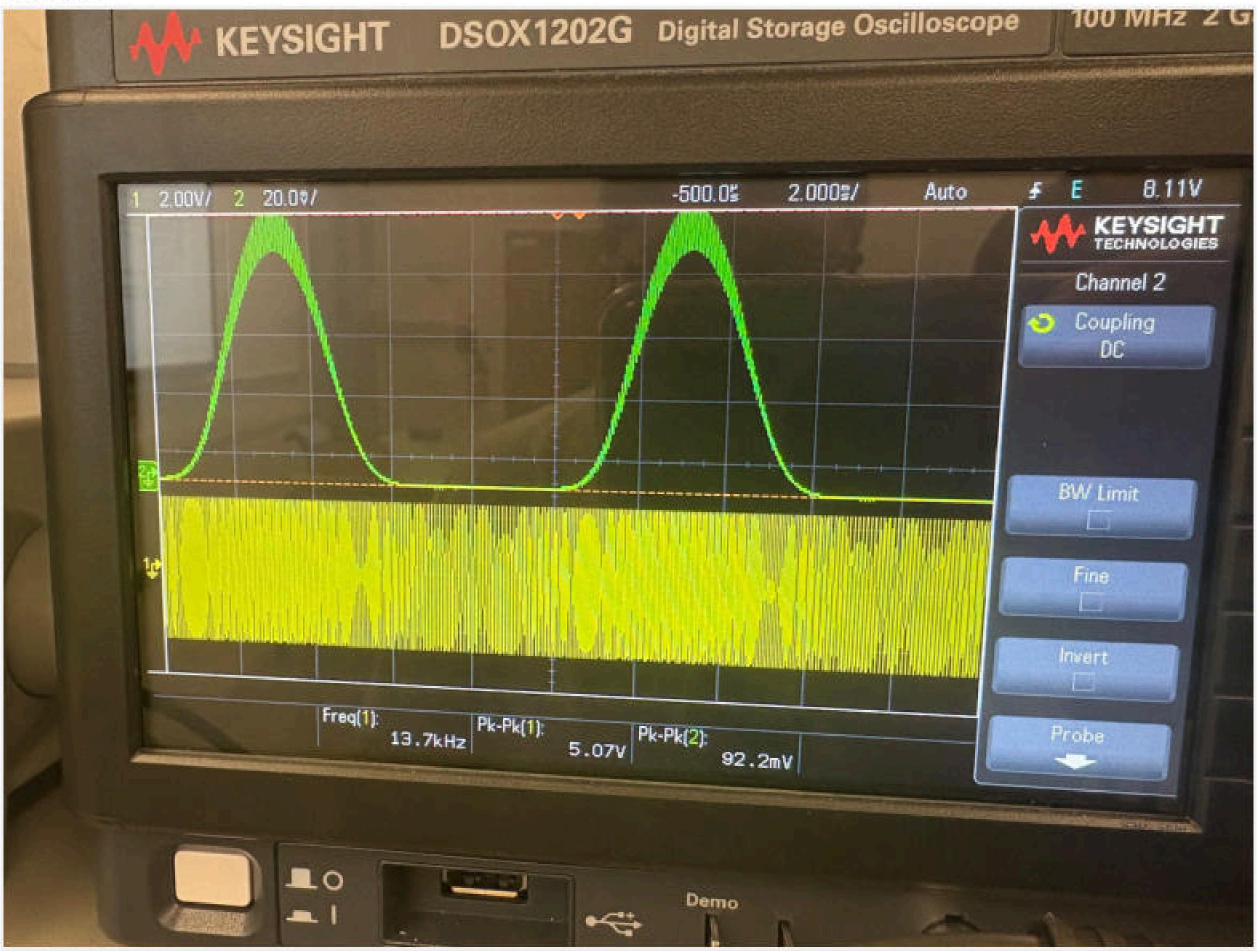
Step4: Use a diode detector: diode, R and C value to rectify and demodulate the pseudo-AM signal out of the filter.

Suggestions: Use a Germanium diode and R – 1kohm connected to a Capacitor substitution box, adjust C value for best output signal i.e., Sine Wave. After this is complete see how the output looks for a FM signal with square wave or triangle wavemodulation.

Sine wave:

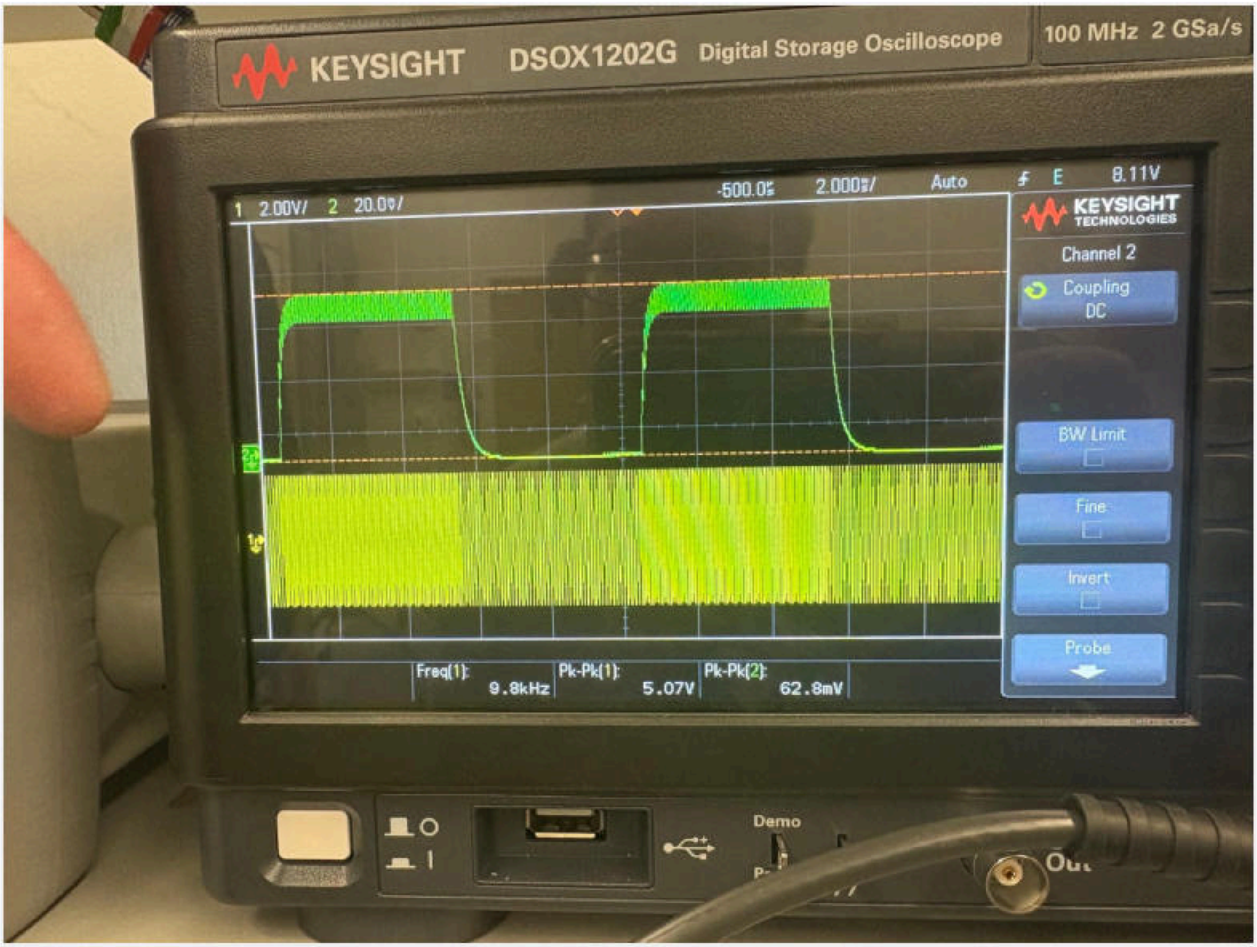


IMG 12



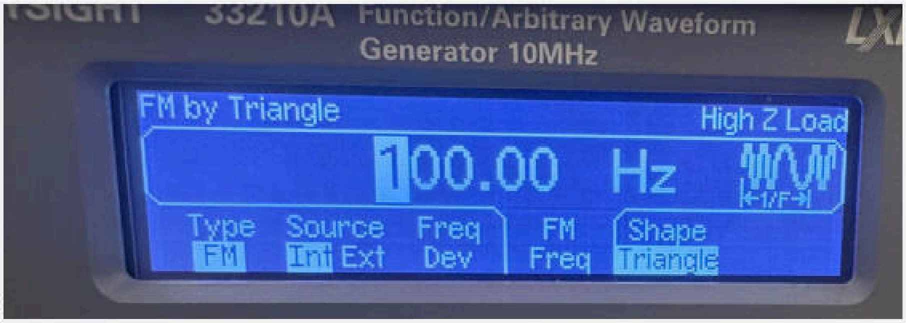
IMG 13

Channel 2 (green) shows the output of the diode detector, which was adjusted to produce a clean sine wave. When the modulation waveform was changed to a triangle wave, the output waveform also changed accordingly. Higher input frequencies produced higher output voltage levels, while lower input frequencies resulted in lower output voltages.

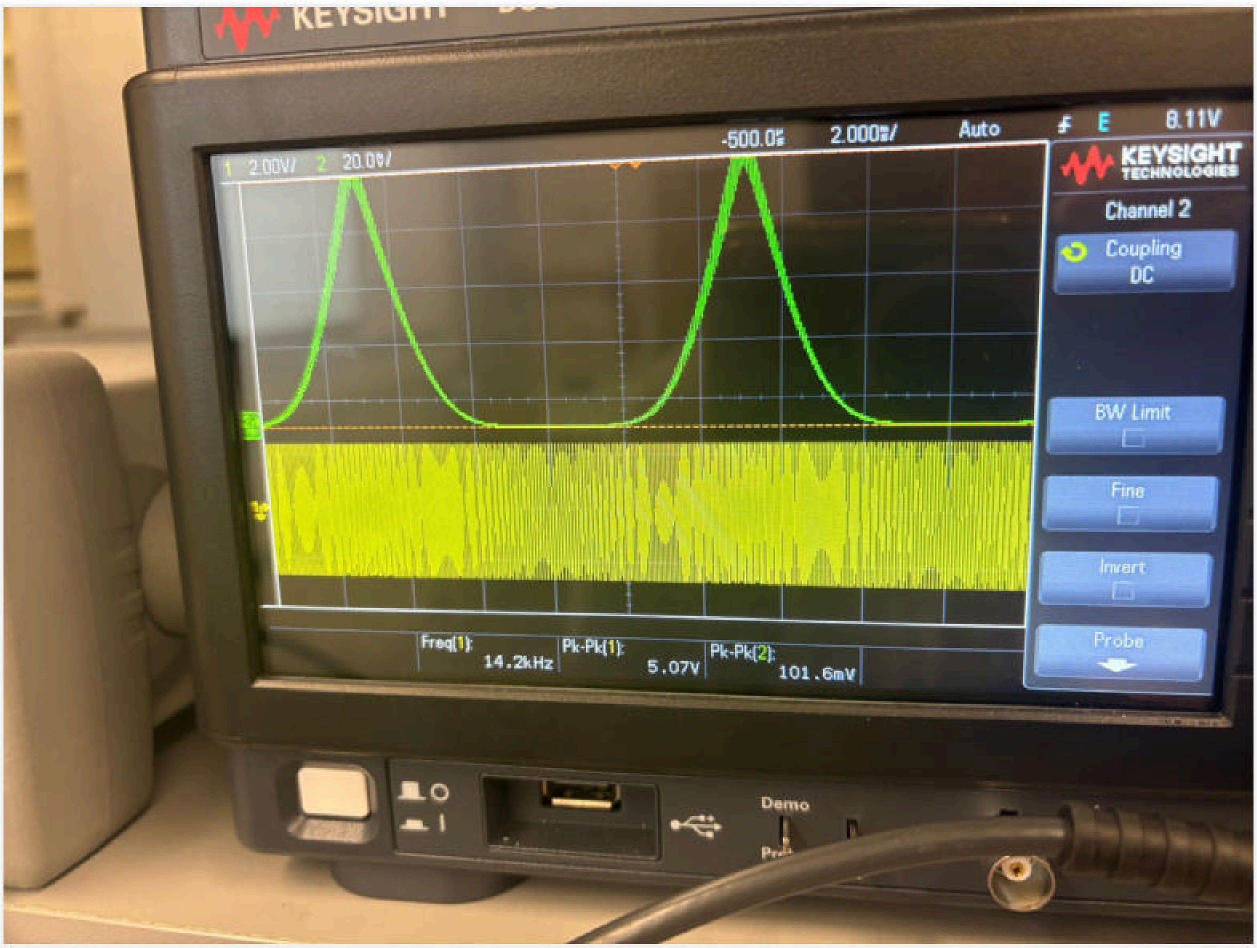


IMG 14.

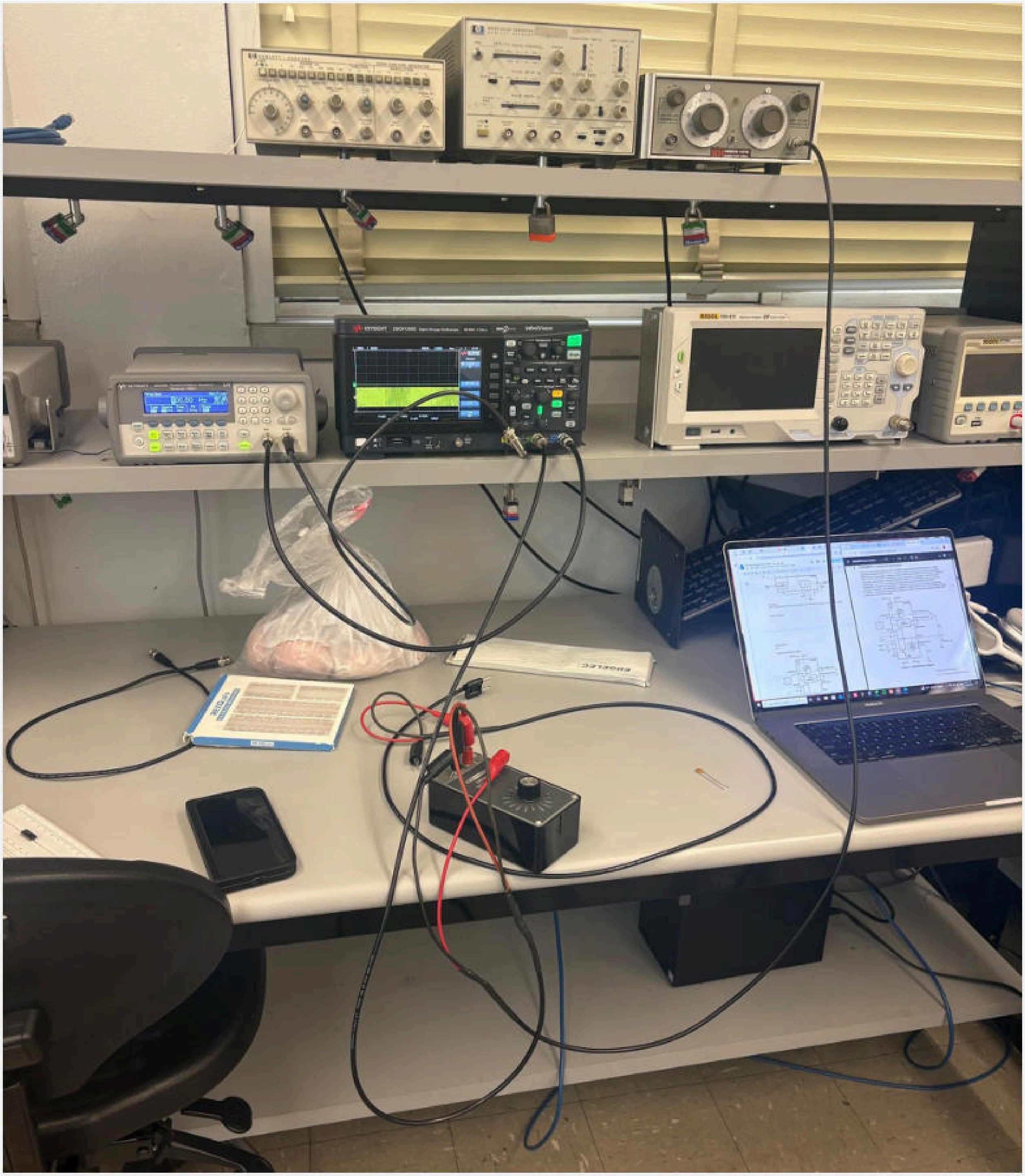
Triangle

:

IMG 15.



IMG 16.

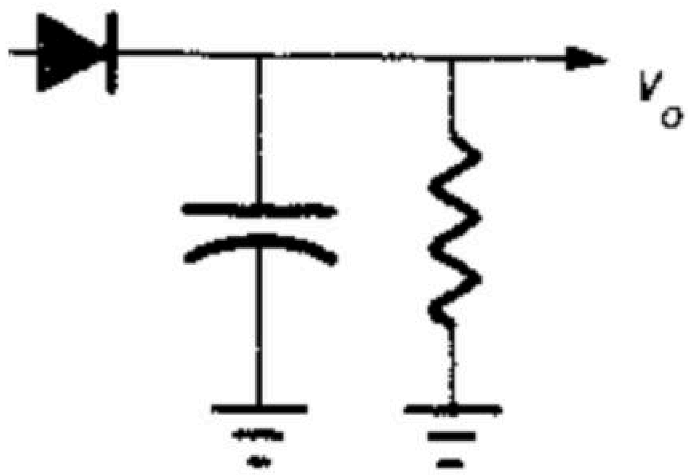


IMG 17. *Set up*

IMG 18 . *The filter*

The modulation frequency was set to 100 Hz with a frequency deviation of 5 kHz using a sinusoidal modulation waveform from the function generator.

Next the filter was configured for FM demodulation using slope detection, following the setup shown below. The Krohn-Hite filter on the bench was used, and the high-pass characteristic (left-side control) was set to approximately 18 kHz, which is slightly above the carrier frequency of 15 kHz. The low-pass control (right-side control) was set to its maximum value of 1 MHz to allow the signal to pass without additional attenuation.



IMG 19 . *Note that this is slightly different then the actual demodulation FM circuit*

Part 2 CD4046 PLL (Phase Locked Loop) FM Demodulator

The PLL is a much more complex FM demodulator as it utilizes a feedback control system based upon phase/frequency changes not voltage as most introductory control systems are configured. As shown in Figure 9-23 from the book a phase detector compares the phase of two signals, the input FM signal and 2. a digital VCO signal, generated internally by the CD4046 IC in our circuit. Uniquely a digital XOR (Exclusive OR) performs this operation see Chapter 10 on PLLs. The phase difference (0-180 degrees) between these two signals is low pass filtered to create a DC signal to control the VCO. If this system is “phase and frequency locked” the phase difference will be zero and VCO will stay at its current frequency. However if the input FM signal increases frequency, the Phase difference increases so the filtered phase difference will drive the VCO to “speedup” or increase its frequency (similar to a car cruise control). Likewise for a decreased FM signal frequency this voltage will decrease. Thus, the filtered voltage signal that drives the VCO follows the input phase/frequency changes and so essentially represents the demodulated signal which is pin 10 of the system. This phase difference (0-180 degrees) is low pass filtered (which extracts a DC value) to drive the VCOwhich tracks the input signal frequency. This phase difference (0-180 degrees) is low pass filtered (which extracts a DC value) to drive the VCO which tracks the input signal frequency,

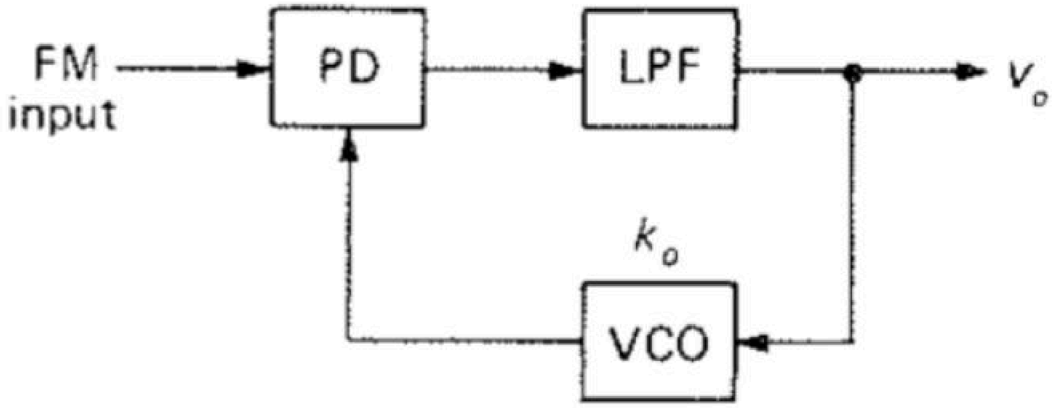
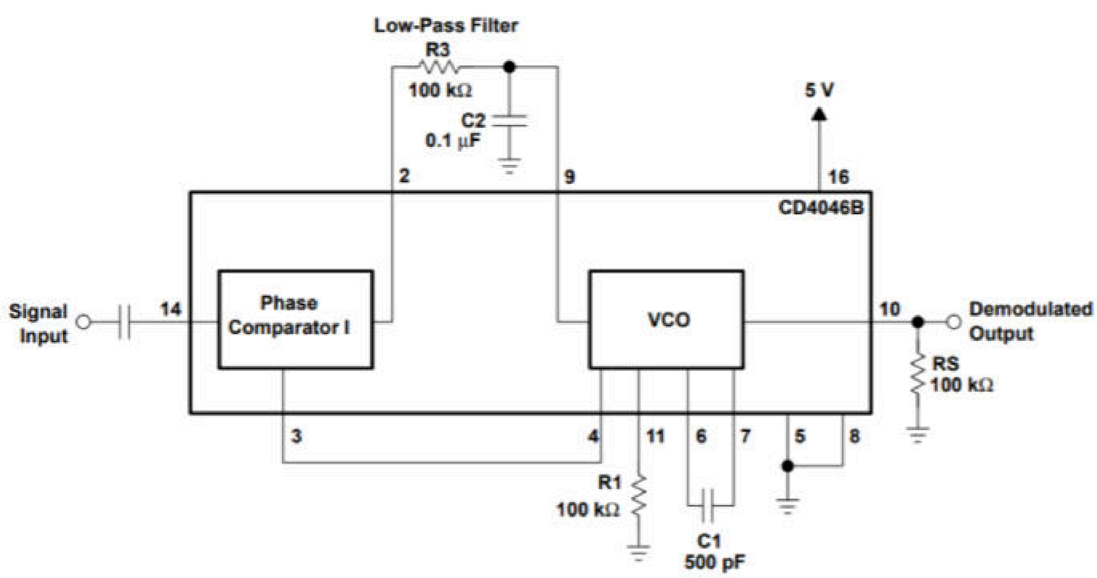


FIGURE 9–23 Phase-locked loop FM detector.

IMG 20.

Step 5: Build the circuit from the datasheet page 17 of http://www.ti.com/lit/an/scha002a/scha002a.pdf The PLL is a much more complex FM demodulator as it utilizes a control system based upon frequency not voltage as most introductory control systems are.



cuit Note(s):

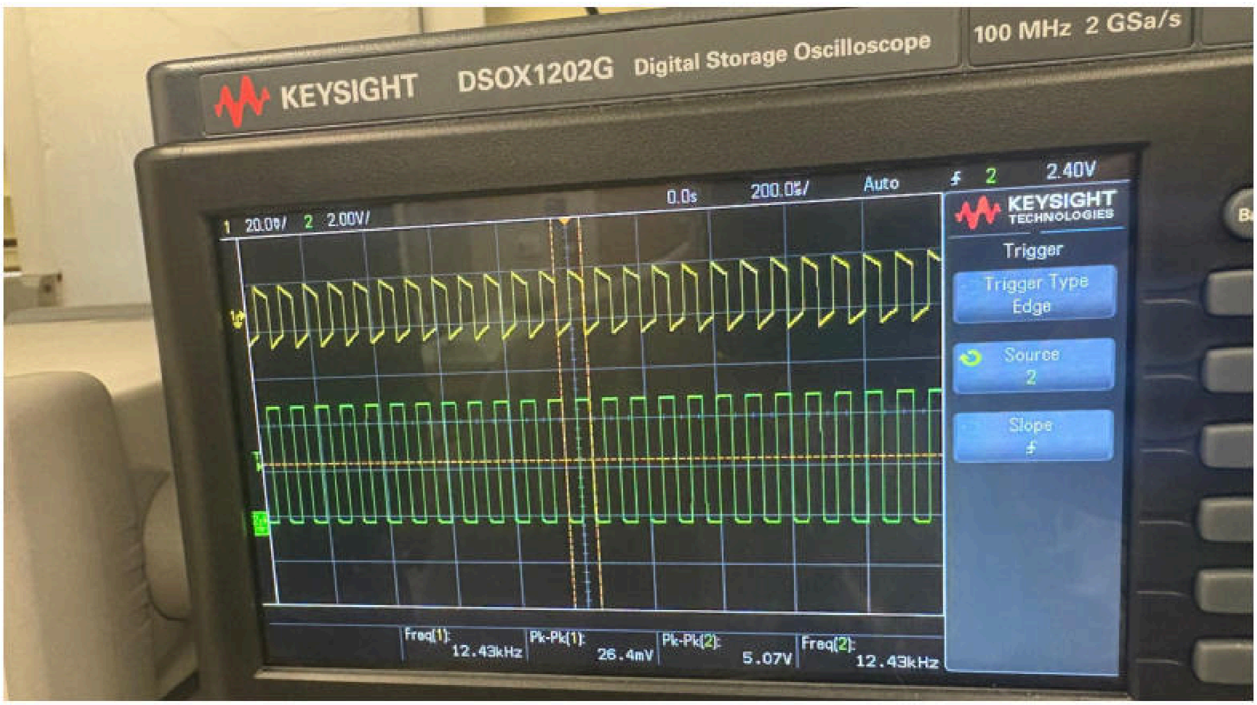
Input C (connected to FM Signal input should be 10 to 100ohms at frequency of interest) for 10 kHz

IMG 21

Circuit Note(s):
a. Input C (connected to FM Signal input should be 10 to 100ohms at frequency of interest) for 10 kHz Xc = 100ohms for 0.159uFd, 1uFd would be 15.9ohms of reactance, a input cap of 0.47uFd or 470nFd will be used for our circuit

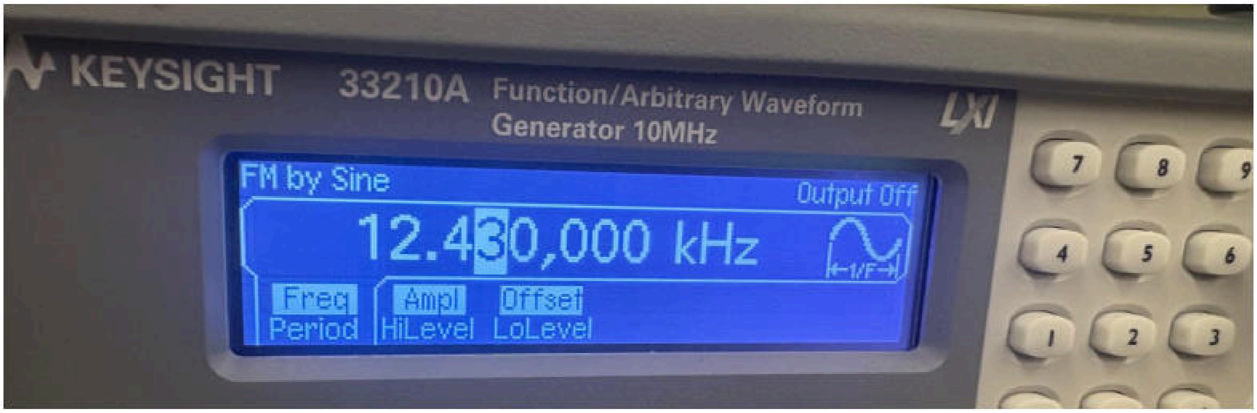
1. With no FM input signal measure the VCO output signal frequency on pins 4 or 3, it should be around 12 kHz or 10 to 20 kHz which is primarily based on C1 at 500pFd. This is the rest frequency of the VCO, review your circuit construction if this is not the cas

The internal and external oscillator frequencies must be set close to each other so that the system can respond correctly and the signal can be observed clearly. If the internal oscillator frequency is too far from the external signal frequency, the system cannot track the changes, resulting in a sampling issue. In this setup, the internal VCO was set to approximately 12.4 kHz.



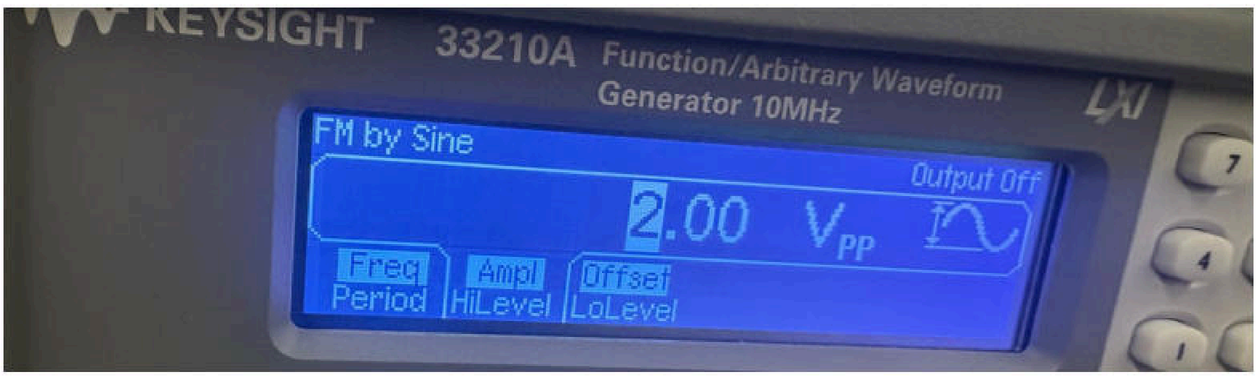
IMG 22.

The internal oscillator was set to approximately 12.43 kHz (green), so the expected sine wave output frequency was about 12.4 kHz, as controlled by the time-base settings. The FM signal from the function generator was applied to the input of the integrated circuit, and a low-pass filter was used as part of a separate circuit box that implements the slope detector for demodulation.



IMG 23 .

2. Create an input signal of 2Vpp (using a bench signal generator) at the rest frequency of the VCO (from part 1) then connect it to FM signal input point and review the pin10 signal of the VCO using a scope.



IMG 24. *input signal of 2 vpp*

3. Add FM modulation to the input signal, with sinusoidal shape and a 1Hz fm value, deviation of 1 kHz). When you view this signal on the scope you should see a noticeable "slinky effect (expansion and contraction) in the signal.

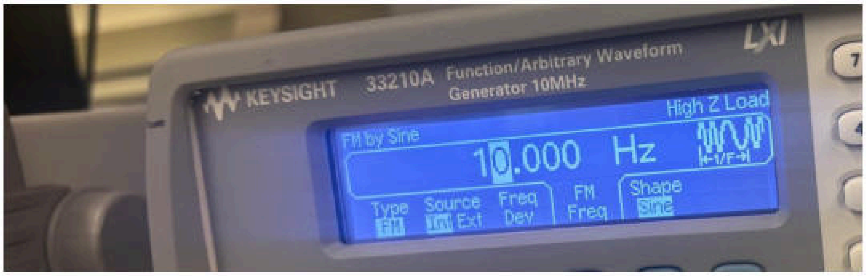
See image 22.

4. Monitor the Demodulated Output on pin 10 and see if you can measure the 1Hz sinusoidal signal recovered from the input FM signal.

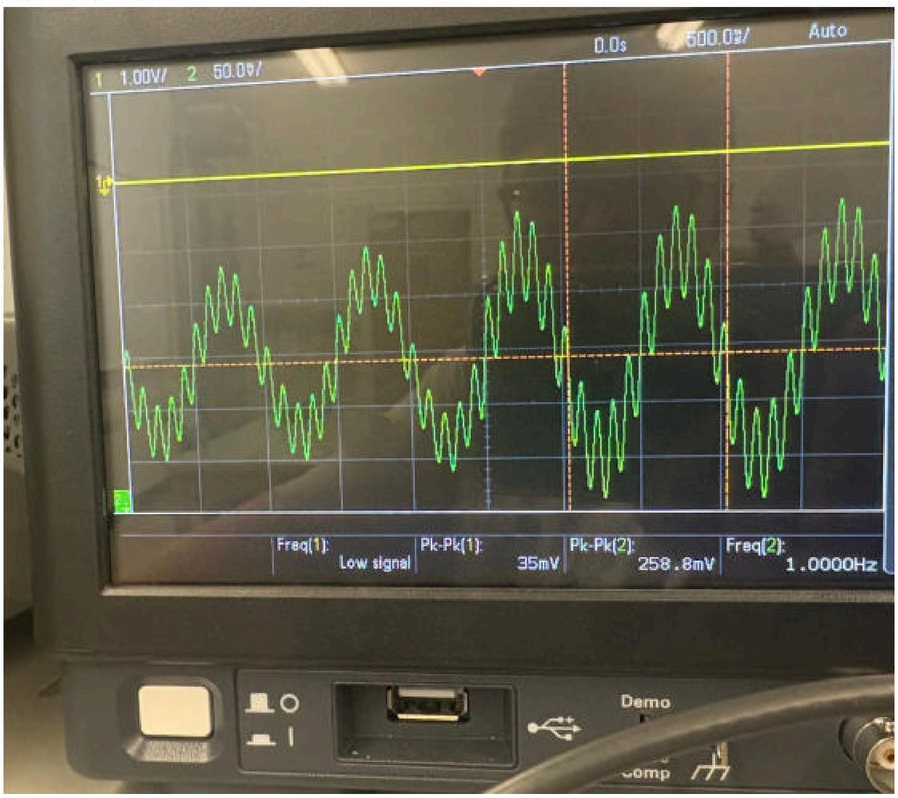
A 1 Hz sinusoidal signal was successfully recovered from the input FM signal. The demodulated output is sinusoidal, which matches the original sin wave. This confirms that when a sine wave is used for modulation, a sine wave is recovered after demodulation.

5. Change the fm value of the FM signal to 10Hz, then 100Hz then up to 1000Hz (1kHz) at some point the signal will vary too much for the PLL and it will lose "lock" and fail to demodulate the input FM signal, when this happens reduce fm in the input signal until it "locks" back in and the signal reappears at the Demodulated Output.

fm value of the FM signal is 10Hz



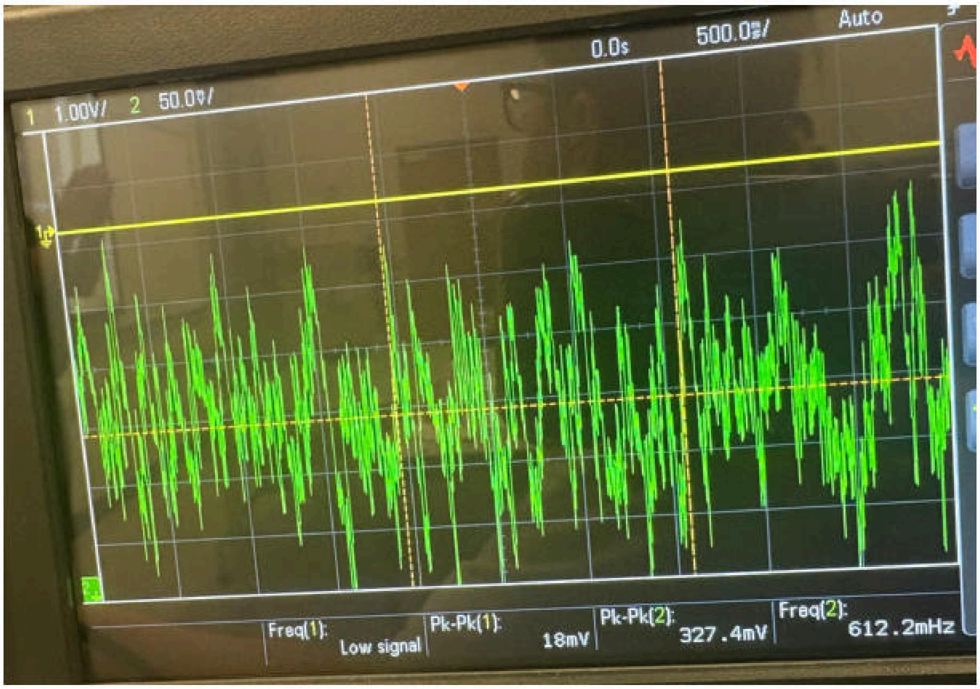
IMG 25. 10 hz



IMG 26

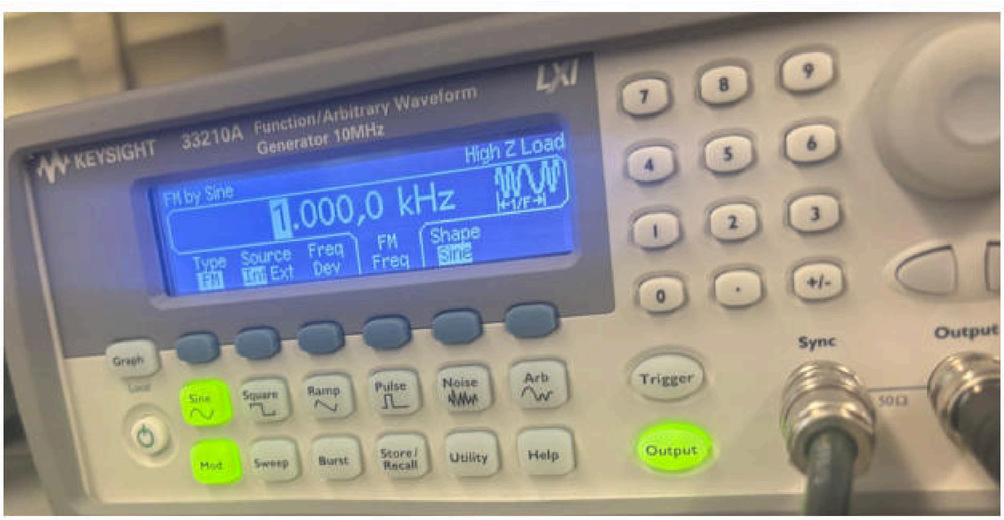
fm value of the FM signal is changed to 100Hz

IMG 27 . *100 hz*

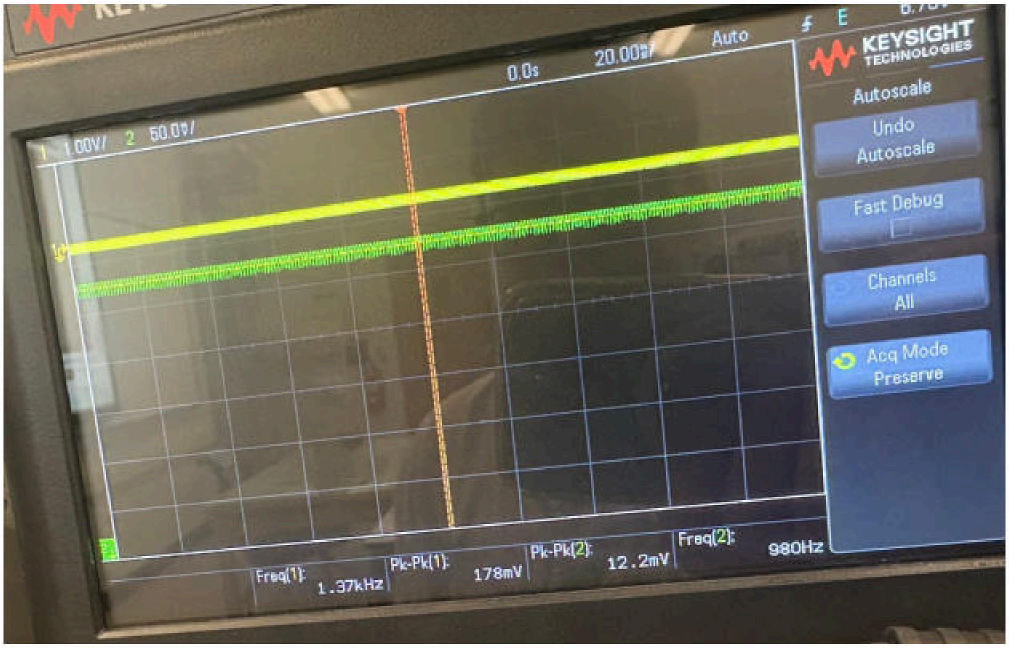


IMG 28

fm value of the FM signal was changed to 1000 hz



IMG 29 . *1000 hz*

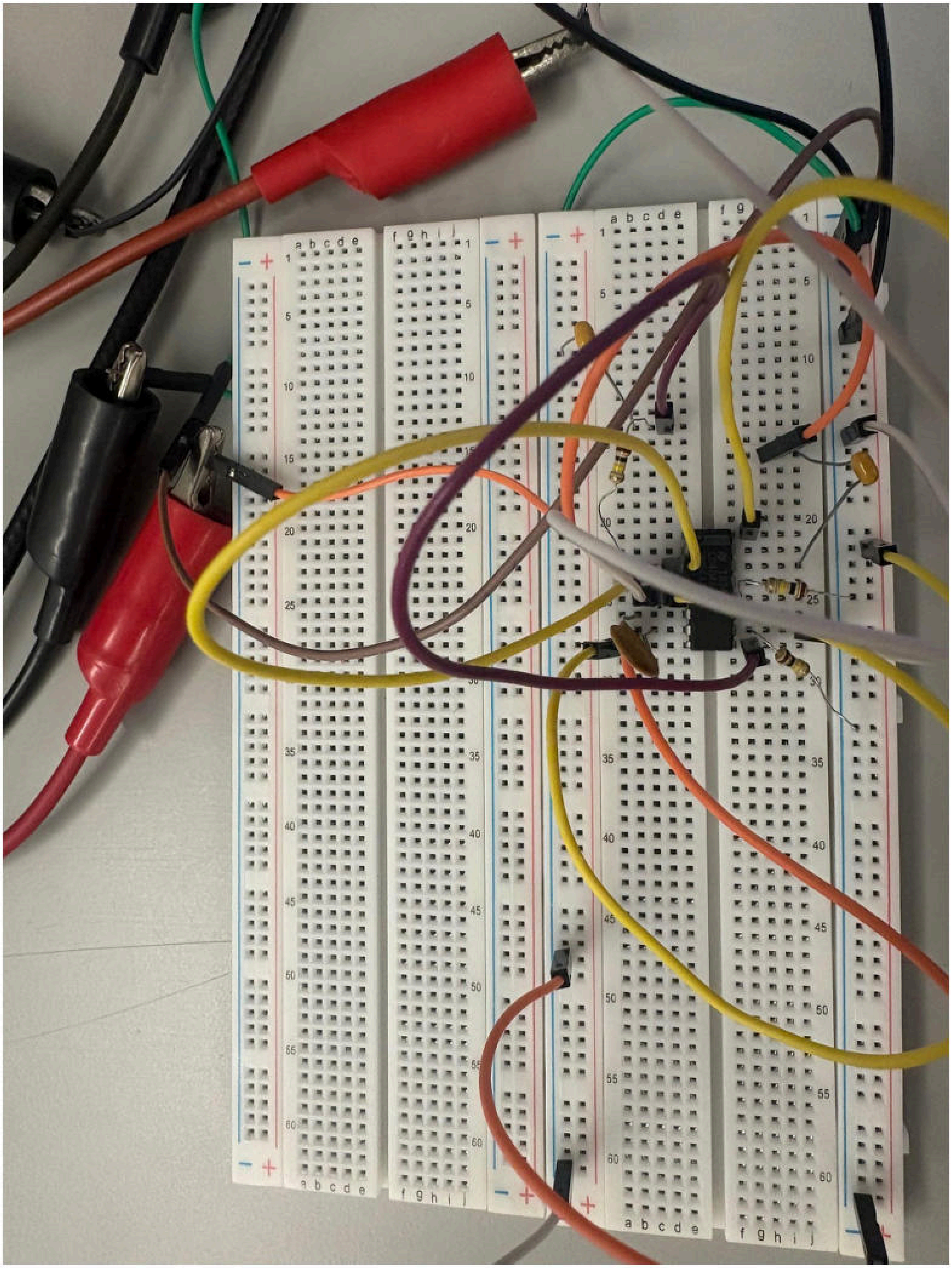


IMG 30 .

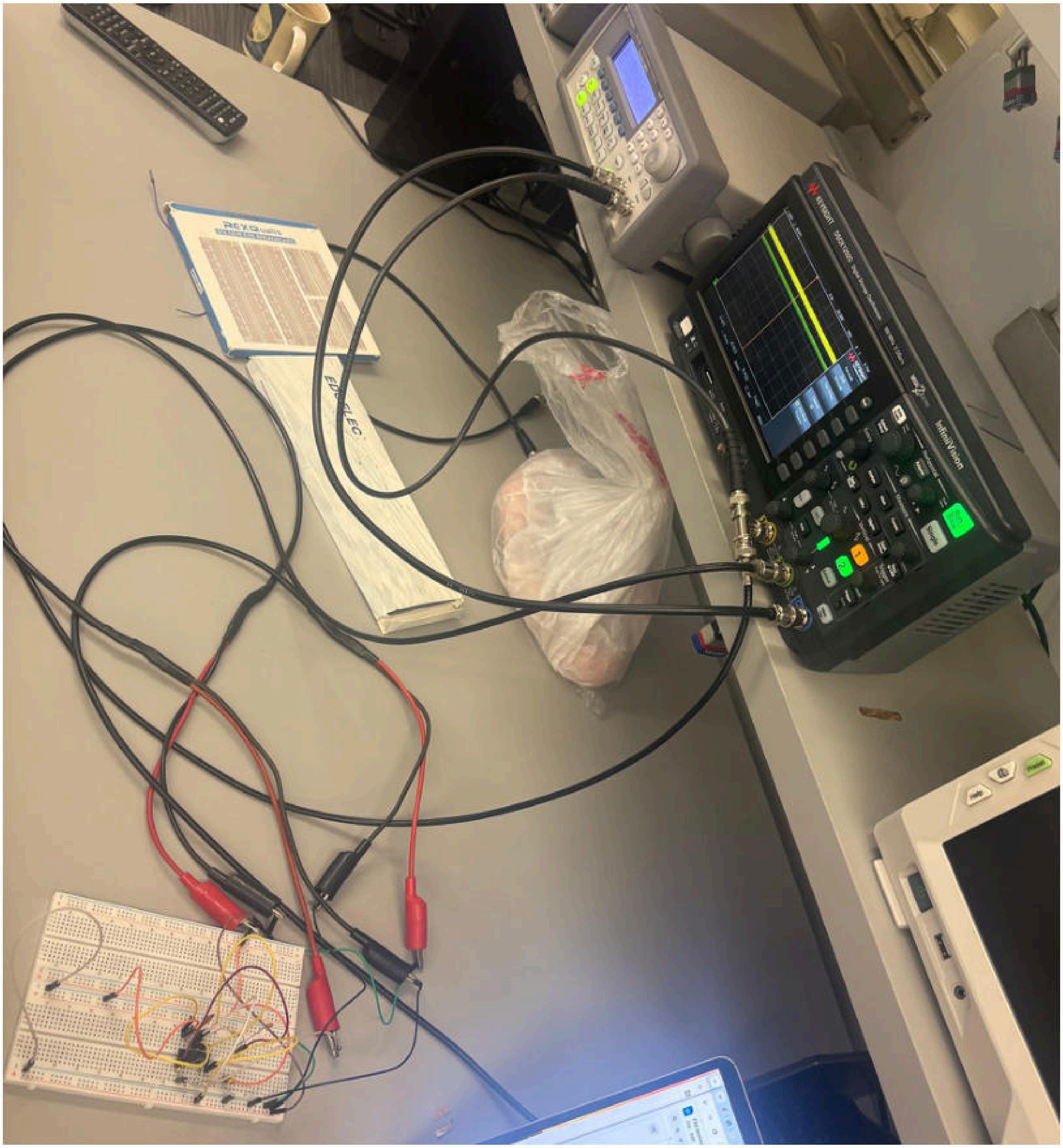
For the Phase-Locked Loop (PLL) to operate correctly, the frequency difference between the input signal and the internal VCO must be within a certain range which is related to the peak frequency deviation. If the input frequency is too far from the VCO frequency, the PLL will unfortunately not be able to lock onto the signal. This is why the internal VCO frequency was measured and adjusted before applying the FM signal.

Practical notes:

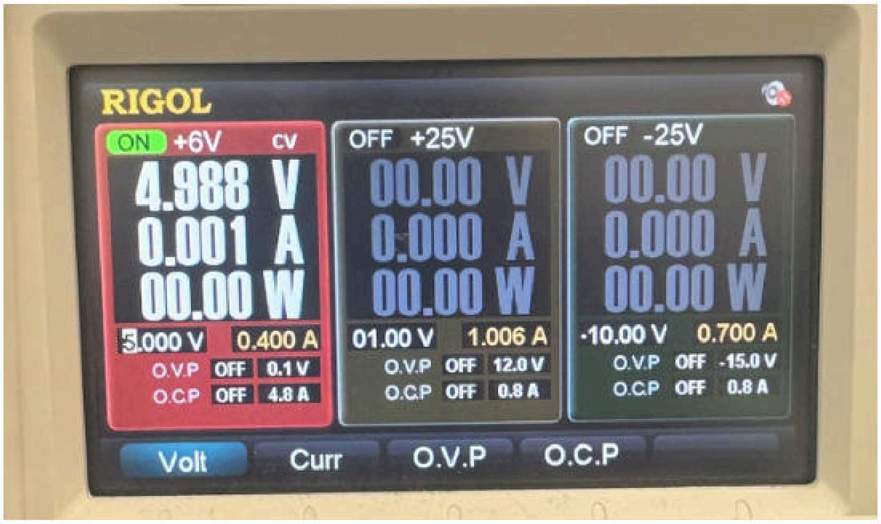
The intermediate frequency should be greater than the peak frequency deviation to allow proper operation. A lower intermediate frequency results in a wider baseband signal, while a higher intermediate frequency makes filtering easier



IMG 31 . *The physical circuit*

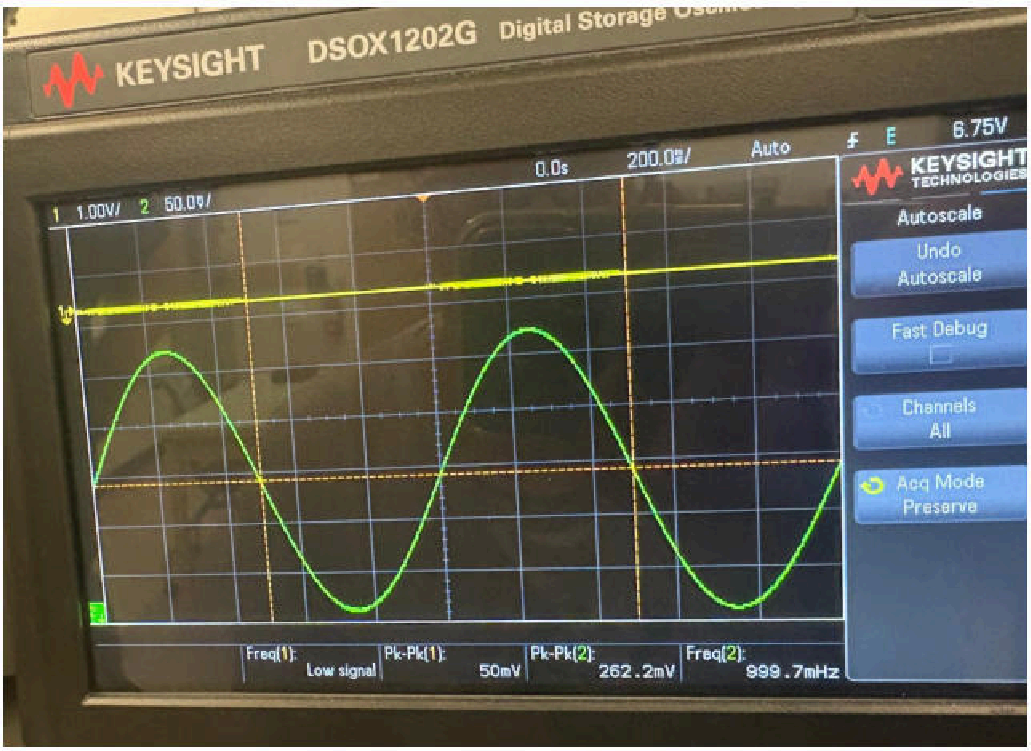


IMG 32 . *System set up*



IMG 33.

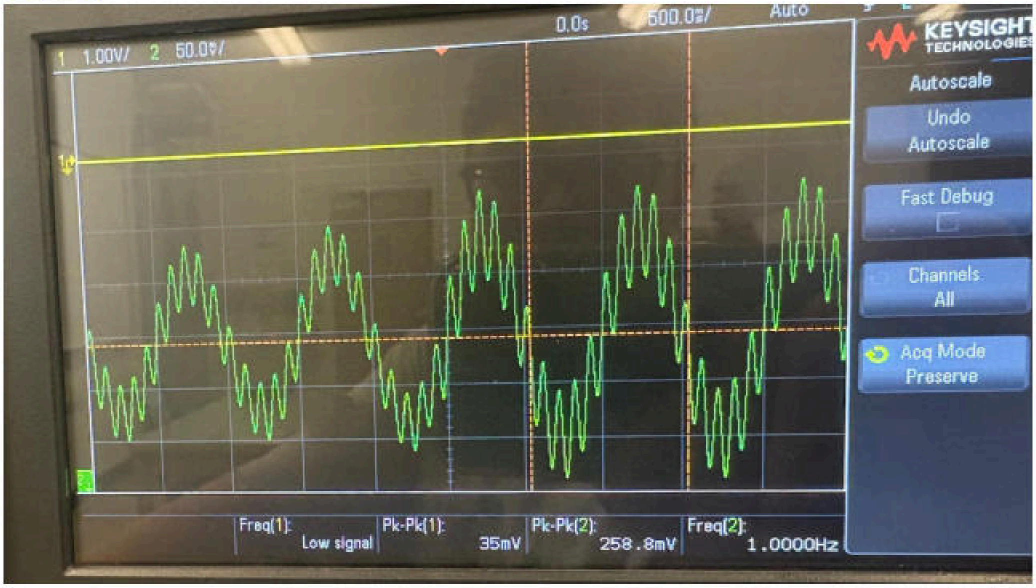
Sine



IMG 34.

6. Return to Step 3 and change the modulation shape of the FM signal to square and see how well the PLL output responds. This type of input FM signal is called FSK or Frequency Shift Keying (keying referring to the on-off nature of control) and is common for simpler digital systems. As before change the FM signal deviation and see if any changes are noticeable.

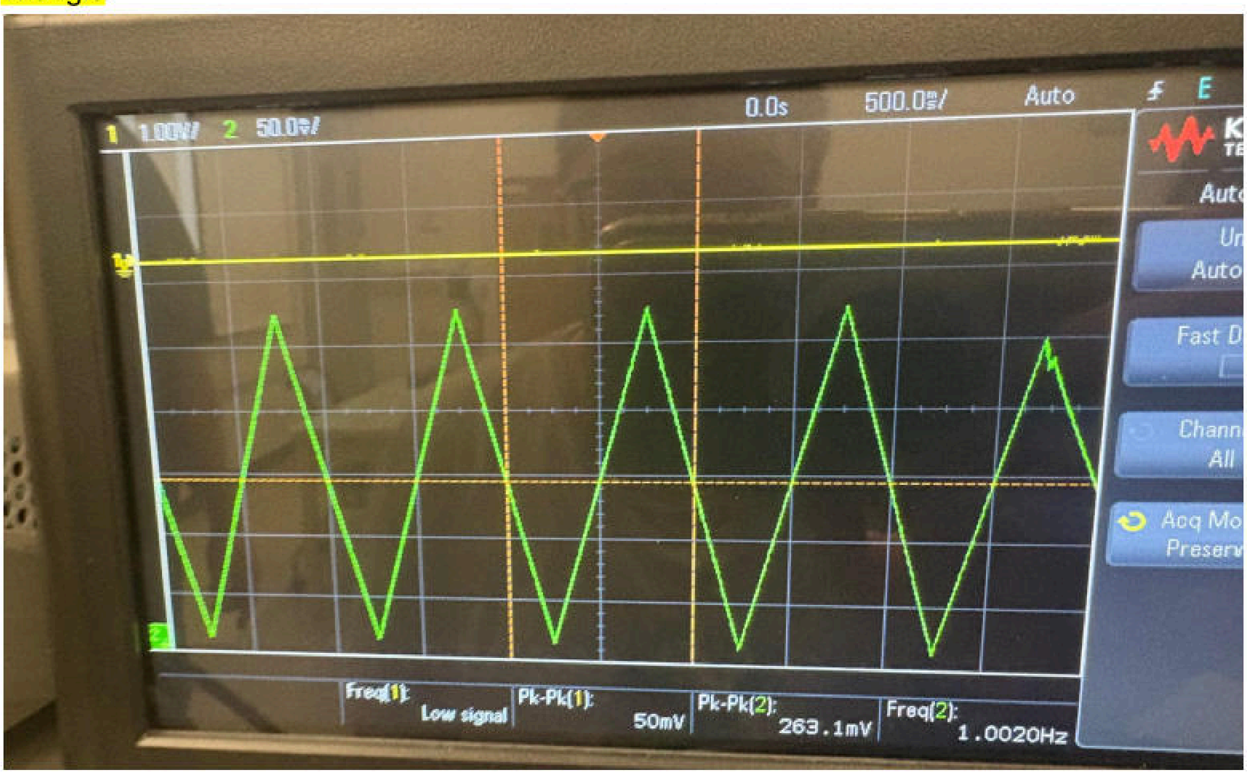
Square



IMG 35.

7. Return to Step 3 and change the modulation shape of the FM signal to a triangle and see how well the PLL output responds. This type of input FM signal is called a Swept FM signal and can be used in RADAR type systems. As before change the FM signal deviation and see if any changes are noticeable.

Triangle



IMG 34. *When we change it to triangle, we see a triangle shape when we demodulate*

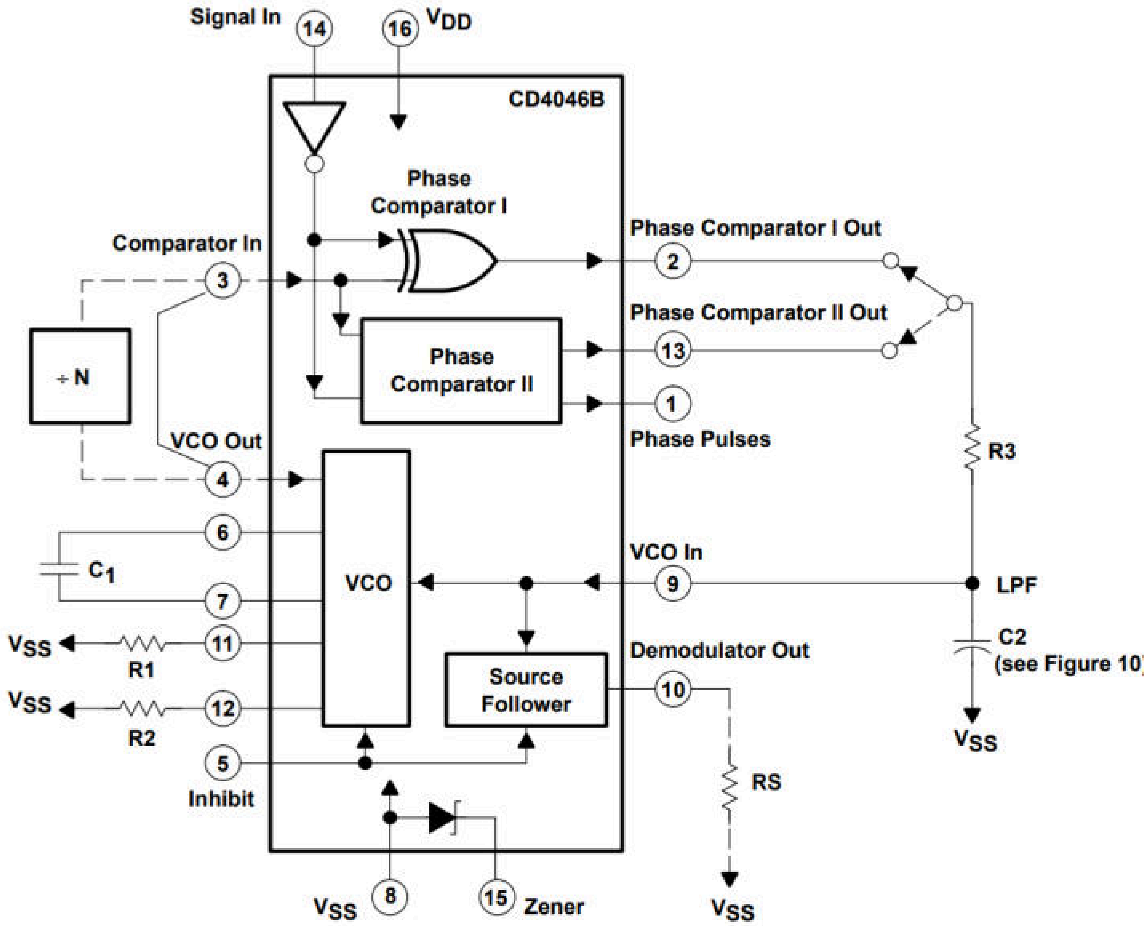


Figure 2. CD4046B Block Diagram

IMG 35.

Conclusion

In this lab we learned how FM signals can be demodulated to recover the original message signal. Unlike AM, FM carries information in changes in frequency, so some extra circuits are needed to convert frequency changes into voltage changes. Using a slope detector, the FM signal was first limited to remove amplitude noise and then passed through a filter so that frequency changes became amplitude changes, which could be detected with a diode. The Phase-Locked Loop (PLL) was also used to demodulate FM by locking onto the input signal and following its frequency changes. This lab helped build a basic understanding of how we demodulate with FM, and how it varies from AM demodulation process.